1. Document History

Document History			
Date	Version	Description	Author
2021-03-23	0.1	First draft version	Edmund Widl (AIT)
2021-05-17	0.2	Add quasi-static thermal network system configuration	Christopher Wild (DTU)
2021-09-01	0.3	Add additional information for the electric network configuration	Tran The Hoang (CEA)
2021-11-19	1.0	Update system configuration to the final implemented benchmark setup	Edmund Widl (AIT)
2022-11-11	1.1	Adapt to ERIGrid 2.0 SC template: • focus on testing / simulation • make template parseable • improved SBD	Edmund Widl (AIT)

2. General description

2.1 System configuration identification

System Configuration Identification	
System configuration ID	Name
MENB-SC	System configuration for the multi-energy network benchmark

2.2 Short description of context

Short Description of Context
Context description
The system configuration described in this document has been developed as part of a reference setup for multi-domain energy network simulations. This reference setup intends to inspire the use of co-simulation for the assessment of multi-energy systems. The level of detail was chosen in a way that allows to apply different modelling approaches for the individual subsystems. For instance, the thermal network could be implemented either in a quasi-static model (series of equilibrium states) or a dynamic model (hydraulic transients). The system configuration resembles part of the sub-urban area of a typical Central European city. The climate is typical for Central Europe with its moderately warm summers and cold winters. Solar irradiation ranges between 800 and 1200 kWh/m². Ambient ground temperature is constant at 8°C.
Key figures
• electrical network: 2 consecutive lines (0.3 km each), connected to external grid • thermal network: 3 main consecutive pipes (supply and return, 0.5 km each), connected to external grid

- power-to-heat facility: 1 heat pump (max. 100 kW_{el}) connected to a thermal tank (100 m^3) feeding into the thermal network
- consumption: 2 consumers, each representing the aggregated loads (electrical and thermal) of an urban quarter and connected to both networks
- generation: 2 PV systems (one of 150 $kW_{el, peak}$ and one of 50 $kW_{el, peak}$)

Keywords

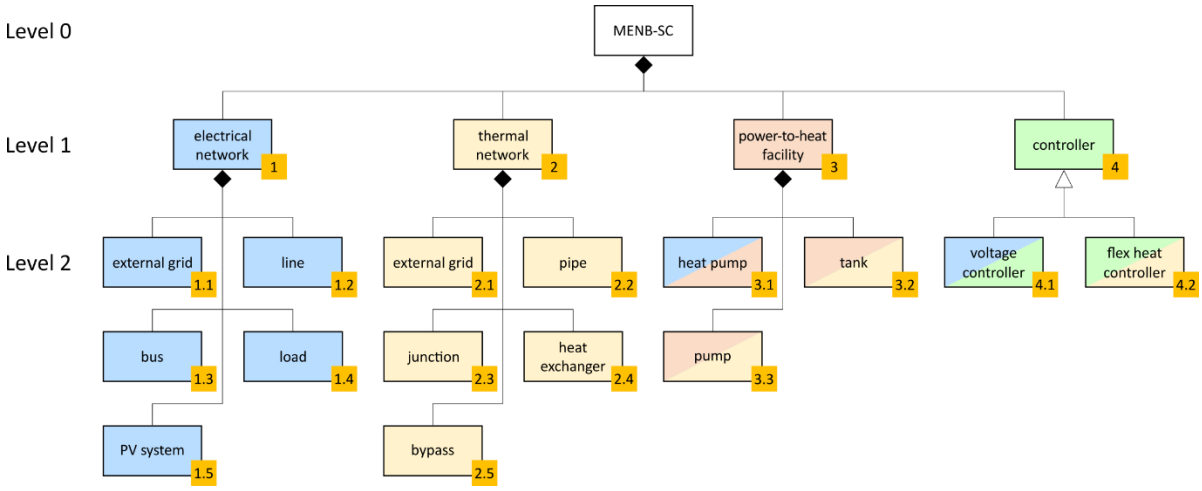
- local consumption of high PV generation
- decentralized heat generation
- heat pumps
- coupled heat and electric networks

Ontology / namespace (optional)

N/A

3. System Breakdown Diagram (SBD)

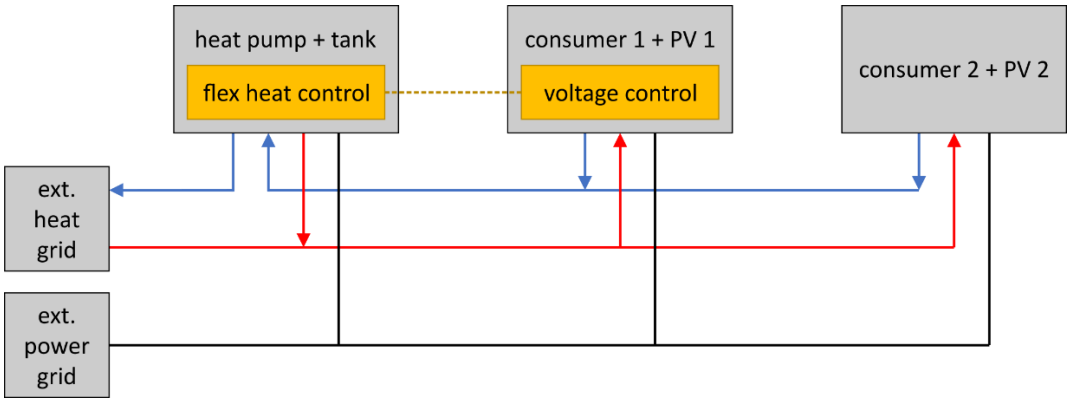
System Breakdown Diagram



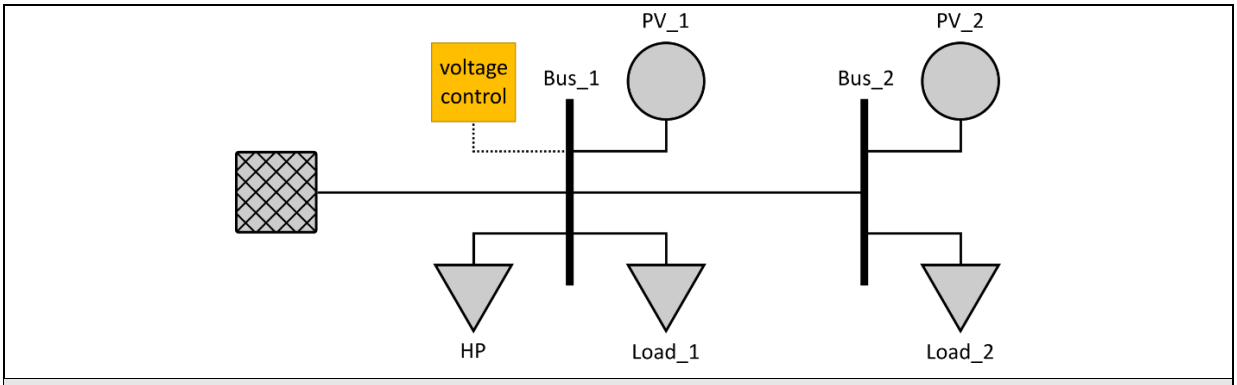
4. Graphical representations of SC

Graphical Representation of System Configuration

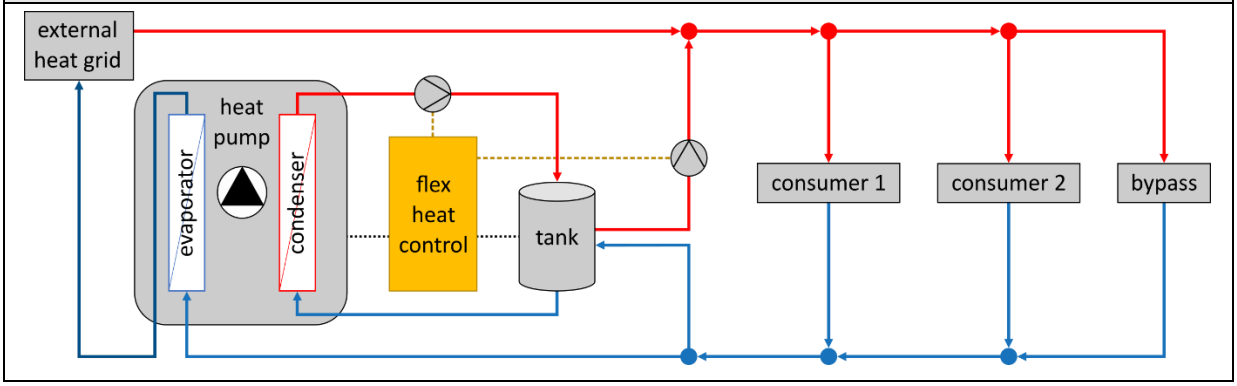
Overview



Electrical Grid (detailed view)



Thermal grid (detailed view):



5. Terminal descriptions

Terminal Description				
Name	Type ID	Type of exchange	Connected component classes	Comment
Heat connector	Heat-Connection	heat flow (water mass flow)	`ExternalElectricGrid`, `Line`, `Busbar`, `Load`, `PVSystem`, `HeatPump`	connects components of the thermal network
Electric power connector	Electric-Connection	electric power flow (electric current)	`ExternalThermalGrid`, `Pipe`, `Junction`, `Bypass`, `HeatExchanger`, `Tank`, `HeatPump`, `Pump`	connects components of the electrical network
Measurement connector	Measure-Connection	measurement exchange (communication signal)	`Bus`, `Tank`, `HeatPump`, `VoltageController`, `FlexHeatController`	connects sensors with controllers via a communication link
Control connector	Ctrl-Connection	control signal exchange (communication signal)	`Pump`, `HeatPump`, `VoltageController`, `FlexHeatController`	connects controllers with actuators via a communication link

6. Component descriptions

6.1 Electrical network

Component description		
Tag in SBD	1	
Level in SBD	1	
Class ID	`ElectricNetwork`	
Parent class	-	
Contained in	`MENB-SC`	
Description	This element represents an electric distribution network. It contains interconnected components with the purpose of delivering electricity to the consumers.	
Number of components in SC	1	
Attributes		
Functionality	The electrical network's actual functionality is determined by its components. This class serves as logical unit containing these components and has itself no functionality.	
Physical characteristics	-	
Terminals		
Terminal ID	Type ID	Description
-	-	-
Instances		
Instance ID	Instance parameterization	
`el_net`	-	

Component description	
Tag in SBD	1.1
Level in SBD	2
Class ID	`ExternalElectricGrid`
Parent class	-
Contained in	`el_net`
Description	This element represents a connection to an external electrical grid supplying the local electrical distribution network.
Number of components in SC	1
Attributes	

Functionality	The external grid controls the voltage and the phase angle of the busbar to which it is connected.	
Physical characteristics	• `f` (float) - Nominal frequency [Hz] • `vn` (float) - Nominal voltage [kV]	
Terminals		
Terminal ID	Type ID	Description
`plug_ext`	`ElectricConnection`	connection point of the electrical distribution network to the external grid
Instances		
Instance ID	Instance parameterization	
`ext_el_grid`	• `f` [Hz]: 220 • `vn` [kV]: 0.4	

Component description		
Tag in SBD	1.2	
Level in SBD	2	
Class ID	`Line`	
Parent class	-	
Contained in	`el_net`	
Description	This element represents a power line of an electrical distribution network.	
Number of components in SC	2	
Attributes		
Functionality	The power lines transport electricity between the busbars of an electrical network.	
Physical characteristics	• `l` (float): line length [km] • `r` (float): resistance [Ω/km] • `x` (float): reactance [Ω/km] • `c` (float): capacitance [nF/km] • `max_i` (float): rated current [kA]	
Terminals		
Terminal ID	Type ID	Description
`plug_line_a`	`ElectricConnection`	connection to a bus of the electrical distribution network on one side of the power line

plug_line_b	ElectricConnection	connection to a bus of the electrical distribution network on other side of the power line
Instances		
Instance ID	Instance parameterization	
line_1	• l [km]: 0.3 • r [Ω/km]: 0.306 • x [Ω/km]: 0.29 • c [nF/km]: 13.2 • max_i [kA]: 0.35	
line_2	• l [km]: 0.3 • r [Ω/km]: 0.306 • x [Ω/km]: 0.29 • c [nF/km]: 13.2 • max_i [kA]: 0.35	

Component description		
Tag in SBD	1.3	
Level in SBD	2	
Class ID	Busbar	
Parent class	-	
Contained in	el_net	
Description	This element represents an electrical busbar used in substations for local power distribution.	
Number of components in SC	2	
Attributes		
Functionality	The electrical busbars connect the components of the electrical network (lines, loads, PV systems).	
Physical characteristics	• vn (float) - nominal voltage [kV]	
Terminals		
Terminal ID	Type ID	Description
plug_bus	ElectricConnection	connection point to other components (lines, loads, PV systems) of the electrical distribution network
meas_voltage_pu	MeasureConnection	voltage measurement (for the voltage controller) [p.u.]
Instances		

Instance ID	Instance parameterization
bus_1	• vn [kV]: 0.4
bus_2	• vn [kV]: 0.4

Component description		
Tag in SBD	1.4	
Level in SBD	2	
Class ID	Load	
Parent class	-	
Contained in	el_net	
Description	This element represents a component of the electrical network that consumes (active) electric power.	
Number of components in SC	3	
Attributes		
Functionality	Each load represents an aggregation of the (active) power consumption of several consumers (e.g., households connected to an LV feeder).	
Physical characteristics	• vn (float): nominal voltage [kV] • p (float): nominal active power consumption [kW]	
Terminals		
Terminal ID	Type ID	Description
plug_load	ElectricConnection	connection to a bus of the electrical distribution network through which power is consumed
Instances		
Instance ID	Instance parameterization	
load_1	• vn [kV]: 0.4 • p [kW]: 120	
load_2	• vn [kV]: 0.4 • p [kW]: 45	

Component description	
Tag in SBD	1.5
Level in SBD	2
Class ID	PVSystem
Parent class	-

Contained in	el_net	
Description	This element represents a PV system comprising several PV panels.	
Number of components in SC	2	
Attributes		
Functionality	The PV systems generates electrical power and supplies it to the electrical distribution network via a busbar connection. Its generation profile is independent of the network conditions.	
Physical characteristics	• P_peak (float): peak production [kW]	
Terminals		
Terminal ID	Type ID	Description
plug_pv	ElectricConnection	connection to a bus of the electrical distribution network through which the generated power is delivered to the electrical network
Instances		
Instance ID	Instance parameterization	
pv_1	• P_peak [kW]: 150	
pv_2	• P_peak [kW]: 50	

6.2 Thermal network

Component description	
Tag in SBD	2
Level in SBD	1
Class ID	ThermalNetwork
Parent class	-
Contained in	MENB-SC
Description	This element represents the thermal network that delivers a mass flow of hot water to satisfy the heat demand of the connected consumers.
Number of components in SC	1
Attributes	
Functionality	The thermal network's actual functionality is mostly determined by its components. This class serves as logical unit containing these components and has itself no functionality.
Physical characteristics	-

Terminals		
Terminal ID	Type ID	Description
-	-	-
Instances		
Instance ID	Instance parameterization	
th_net	–	

Component description		
Tag in SBD	2.1	
Level in SBD	2	
Class ID	ExternalThermalGrid	
Parent class	-	
Contained in	th_net	
Description	This element represents a connection to an external higher-level thermal network supplying the local thermal network.	
Number of components in SC	1	
Attributes		
Functionality	The external thermal network acts as a pressure source with a prescribed supply temperature. It is considered an ideal heating unit without any constraints regarding mass flow or ramp rates.	
Physical characteristics	• p_ext (float): pressure of the external thermal network [bar] • T_supply_ext (float): nominal supply temperature [°C]	
Terminals		
Terminal ID	Type ID	Description
flange_supply	HeatConnection	supply line connection to external thermal grid
flange_return	HeatConnection	return line connection to external thermal grid
Instances		
Instance ID	Instance parameterization	
ext_th_grid	• p_ext [bar]: 6 • T_supply_ext [°C]: 75	

Component description	
Tag in SBD	2.2
Level in SBD	2

Class ID	Pipe	
Parent class	-	
Contained in	th_net	
Description	This element represents a straight pipe with constant cross section used to transport water within the thermal network.	
Number of components in SC	14 (7x supply, 7x return)	
Attributes		
Functionality	Determines the transport effects (pressure drop, flow delays, etc.) for a mass flow based on hydraulic principles. Depending on the insulation of the pipe and the surrounding temperature, the temperature of the water transported through the pipe changes.	
Physical characteristics	• l (float): pipe length [km] • d (float): pipe diameter [m] • alpha (float): heat transfer coefficient [$Wm^{-2}K^{-1}$] • k (float): roughness of pipe material [mm]	
Terminals		
Terminal ID	Type ID	Description
flange_a	HeatConnection	connection to another component of the thermal network on one side of the pipe
flange_b	HeatConnection	connection to another component of the thermal network on other side of the pipe
Instances		
Instance ID	Instance parameterization	
pipe_1_supply	• l [km]: 0.5 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01	
pipe_1_return	• l [km]: 0.5 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01	
pipe_2_supply	• l [km]: 0.5 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01	
pipe_2_return	• l [km]: 0.5 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01	

pipe_4_supply	• l [km]: 0.5 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_4_return	• l [km]: 0.5 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_tank_supply	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_tank_return	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_3_supply	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_3_return	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_5_supply	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_5_return	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_6_supply	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01
pipe_6_return	• l [km]: 0.01 • d [m]: 0.1 • alpha [$Wm^{-2}K^{-1}$]: 1.5 • k [mm]: 0.01

Component description	
Tag in SBD	2.3

Level in SBD	2	
Class ID	`Junction`	
Parent class	-	
Contained in	`th_net`	
Description	This element represents a tee junction for connecting three pipes.	
Number of components in SC	6	
Attributes		
Functionality	Used either for splitting the mass flow from a single pipe into separate mass flows of two pipes of or joining mass flows from two separate pipes into a single mass flow for one pipe. Splitting/joining follows basic hydraulic rules (e.g., mass conservation).	
Physical characteristics	-	
Terminals		
Terminal ID	Type ID	Description
`flange_1`	`HeatConnection`	connection to pipe 1
`flange_2`	`HeatConnection`	connection to pipe 2
`flange_3`	`HeatConnection`	connection to pipe 3
Instances		
Instance ID	Instance parameterization	
`junction_1_supply`	(branch-off from supply line to consumers)	
`junction_2_supply`	(branch-off from supply line to consumers)	
`junction_1_return`	branch-off from return line to consumers	
`junction_2_return`	branch-off from return line to consumers	
`junction_tank_supply`	branch-off from supply line to tank	
`junction_tank_return`	branch-off from return line to tank	

Component description	
Tag in SBD	2.4
Level in SBD	2
Class ID	`HeatExchanger`
Parent class	-
Contained in	`th_net`
Description	This element represents a heat exchanger acting as an aggregated heat consumer in the thermal network.
Number of components in SC	2

Attributes		
Functionality	The heat exchanger is fed with hot water from the thermal network's supply line. The mass flow through the heat exchanger is adjusted to reach a pre-defined return temperature, depending on the current heat demand from the aggregated consumer.	
Physical characteristics	`T_return_target` (float): set-point for return temperature [°C]	
Terminals		
Terminal ID	Type ID	Description
`flange_supply`	`HeatConnection`	inlet of heat exchanger from supply line
`flange_return`	`HeatConnection`	outlet of heat exchanger to return line
Instances		
Instance ID	Instance parameterization	
`hex_1`	• `T_return_target` [°C]: 40	
`hex_2`	• `T_return_target` [°C]: 40	

Component description		
Tag in SBD	2.5	
Level in SBD	2	
Class ID	`Bypass`	
Parent class	-	
Contained in	`th_net`	
Description	This element represents the bypass of the thermal network.	
Number of components in SC	1	
Attributes		
Functionality	Ideal valve that allows to regulate the water mass flow from the supply line to the return line independently of the consumers.	
Physical characteristics	-	
Terminals		
Terminal ID	Type ID	Description
`flange_supply`	`HeatConnection`	inlet of bypass from supply line
`flange_return`	`HeatConnection`	outlet of bypass to return line
Instances		
Instance ID	Instance parameterization	
`bypass`	_	

6.3 Power-to-heat facility

Component description		
Tag in SBD	3	
Level in SBD	1	
Class ID	`PowerToHeatFacility`	
Parent class	-	
Contained in	`MENB-SC`	
Description	This element represents a power-to-heat facility comprising a heat pump connected to a thermal tank, which feeds into the thermal network's supply line to support its operation. By consuming local excess PV generation, the power-to-heat facility can be used at the same time to improve the stability of the electrical network and support the supply of the thermal network. Its components are actuated by the flex heat controller.	
Number of components in SC	1	
Attributes		
Functionality	The power-to-heat facility's actual functionality is determined by its components. This class serves as logical unit containing these components and has itself no functionality.	
Physical characteristics	-	
Terminals		
Terminal ID	Type ID	Description
–	–	-
Instances		
Instance ID	Instance parameterization	
`p2h`	–	

Component description	
Tag in SBD	3.1
Level in SBD	2
Class ID	`HeatPump`
Parent class	-
Contained in	`p2h`
Description	This element represents a heat pump with constant condenser output temperature.

Number of components in SC	1	
Attributes		
Functionality	The source for the evaporator is the thermal network's return line. The condenser's output is fed to the thermal storage tank.	
Physical characteristics	• `P_rated_max` (float): electrical power rating of compressor [kW_{el}] • `P_rated_min` (float): minimal electrical power consumption for operating the compressor [kW_{el}] • `P_0` (float): electrical stand-by power consumption [kW_{el}] • `T_evap_min` (float): minimal evaporator outlet temperature [°C] • `T_cond_max` (float): maximum condenser outlet temperature [°C] • `T_cond_target` (float): condenser outlet temperature setpoint [°C] • `eta_sys` (float): ratio between work provided by the pump and available thermodynamic work [-] • `eta_comp` (float): compressor efficiency [-] • `lambda_comp` (float): compressor time constant [s^{-1}]	
Terminals		
Terminal ID	Type ID	Description
`plug_load`	`ElectricConnection`	connection to a bus of the electrical distribution network through which power is consumed
`flange_evap_in`	`HeatConnection`	inlet of heat pump evaporator from return line
`flange_evap_out`	`HeatConnection`	outlet of heat pump evaporator to return line
`flange_cond_in`	`HeatConnection`	inlet of heat pump condenser from tank
`flange_cond_out`	`HeatConnection`	outlet of heat pump condenser to tank
`meas_p_el`	`MeasureConnection`	measurement of electrical power consumption (for flex heat controller) [kW_{el}]
Instances		
Instance ID	Instance parameterization	
`hp`	• `P_rated` [kW_{el}]: 100	

	• P_0 [kW_{el}]: 0.3 • T_evap_min [°C]: 20 • T_cond_max [°C]: 85 • T_cond_target [°C]: 75 • eta_sys [-]: 0.5 • eta_comp [-]: 0.7 • lambda_comp [s^{-1}]: 0.2

Component description		
Tag in SBD	3.2	
Level in SBD	2	
Class ID	Tank	
Parent class	-	
Contained in	p2h	
Description	This element represents a stratified hot water storage tank.	
Number of components in SC	1	
Attributes		
Functionality	The tank has two hydraulic loops (also see pumps below): • cold water is drawn from the bottom of the tank, heated up by the heat pump and fed back at the top of the tank • hot water is drawn from the top of the tank and fed to the heating network's supply line; simultaneously an equal amount of cold water is drawn from the heating network's return line to refill the tank	
Physical characteristics	• V (float): tank volume [m^3] • h (float): height of tank (without insulation) [m] • d (float): thickness of insulation [m] • k (float): specific heat conductivity of insulation [$Wm^{-1}K^{-1}$] • N (int): number of volume segments [-]	
Terminals		
Terminal ID	Type ID	Description
flange_tank_supply	HeatConnection	hot water outlet of tank to supply line
flange_tank_return	HeatConnection	cold water inlet from return line to tank
flange_hp_out	HeatConnection	cold water outlet from tank to heat pump condenser
flange_hp_in	HeatConnection	hot water inlet from heat pump condenser to tank

meas_temp	MeasureConnection	tank temperature measurement (for flex heat controller) [°C]
Instances		
Instance ID	Instance parameterization	
tank	• V [m^3]: 100 • h [m]: 9.2 • d [m]: 0.1 • k [$Wm^{-1}K^{-1}$]: 0.03 • N [-]: 10	

Component description		
Tag in SBD	3.3	
Level in SBD	2	
Class ID	Pump	
Parent class	-	
Contained in	p2h	
Description	This element represents a water pump with prescribed mass flow rate.	
Number of components in SC	2	
Attributes		
Functionality	Ideal water pump that generates a mass flow according to the setpoint. Moves the water in the following loops of the power-to-heat facility: • condenser loop: pump hot water from the heat pump condenser to the tank • network support loop: pump hot water from the tank to the supply line	
Physical characteristics	-	
Terminals		
Terminal ID	Type ID	Description
flange_in	HeatConnection	pump inlet
flange_out	HeatConnection	pump outlet
setpoint_mflow	CtrlConnection	set-point for pump mass flow (from flex heat controller)
Instances		
Instance ID	Instance parameterization	

pump_cond	pump for tank charging loop
pump_tank	pump for tank discharging loop

6.4 Controls

Component description		
Tag in SBD	4	
Level in SBD	1	
Class ID	Controller	
Parent class	-	
Contained in	MENB-SC	
Description	This element serves as parent class for concrete controller implementations.	
Number of components in SC	-	
Attributes		
Functionality	• Controllers govern the operation of the system.	
Physical characteristics	-	
Terminals		
Terminal ID	Type ID	Description
–	–	-
Instances		
Instance ID	Instance parameterization	
-	-	

Component description	
Tag in SBD	4.1
Level in SBD	2
Class ID	VoltageController
Parent class	Controller
Contained in	-
Description	This element represents a voltage controller in the electrical distribution network.
Number of components in SC	1
Attributes	

Functionality	• This controller monitors the voltage at `bus_1` and proposes a power consumption setpoint for the heat pump (controllable/flexible load) to keep the voltage within acceptable limits.
Physical characteristics	• `delta_vm_up` (float): upper threshold for turning off the heat pump [p.u.] • `delta_vm_low_hp_on` (float): lower threshold for turning on the heat pump [p.u.] • `delta_vm_low_hp_off` (float): lower threshold for turning off the heat pump [p.u.] • `delta_vm_deadband` (float): deadband size [p.u.] • `hp_p_el_mw_min` (float): minimum operating point (minimal allowed power consumption) of heat pump [MWe] • `k_p` (float): the controller's proportional term [-]

Terminals		
Terminal ID	Type ID	Description
`meas_voltage_pu`	`MeasureConnection`	voltage measurement at bus_1 [p.u.]
`setpoint_hp_p_el`	`CtrlConnection`	proposed heat pump setpoint for electrical consumption [MW_{el}]

Instances	
Instance ID	Instance parameterization
`voltage_ctrl`	• `delta_vm_up` [p.u.]: 0.1 • `delta_vm_low_hp_on` [p.u.]: – 0.1 • `delta_vm_low_hp_off` [p.u.]: – 0.08 • `delta_vm_deadband` [p.u.]: 0.03 • `hp_p_el_mw_min` [kW_{el}]: 35 • `k_p` [-]: 0.15

Component description	
Tag in SBD	4.2
Level in SBD	2
Class ID	`FlexHeatController`
Parent class	`Controller`
Contained in	-
Description	This element represents a dedicated controller for operating the power-to-heat facility.
Number of components in SC	1

Attributes		
Functionality	• The flex heat controller decides whether the heat supply is covered entirely through the external grid or whether the power-to-heat facility supports by discharging the tank. If required, the heat pump is used to charge the tank, taking into account the power consumption threshold from the voltage controller (i.e., the power consumption never exceeds the setpoint, but may be less).	
Physical characteristics	• `T_tank_max` (float): maximum tank temperature [°C] • `T_tank_min` (float): minimum tank temperature [°C] • `mdot_tank_out_setpoint` (float): setpoint for tank discharge mass flow rate [kg/s]	
Terminals		
Terminal ID	Type ID	Description
`meas_tank_temp`	`MeasureConnection`	measurement of tank temperature [°C]
`meas_hp_p_el`	`MeasureConnection`	measurement of electrical power consumption of heat pump [kW_{el}]
`setpoint_hp_p_el`	`CtrlConnection`	setpoint for power consumption of heat pump from voltage controller [MW_{el}]
`setpoint_tank_mflow`	`CtrlConnection`	setpoint for tank discharge mass flow [kg/s]
`setpoint_cond_mflow`	`CtrlConnection`	setpoint for heat pump condenser mass flow [kg/s]
Instances		
Instance ID	Instance parameterization	
`flex_heat_ctrl`	• `T_tank_max` [°C]: 72 • `T_tank_min` [°C]: 65	

7. Connection points

Connection points			
Connection from terminal	Connection from terminal	Connection from terminal	Connection from terminal
`ext_el_grid`	`plug_ext`	`line_1`	`plug_line_a`
`bus_1`	`plug_bus`	`line_1`	`plug_line_b`
`bus_1`	`plug_bus`	`line_2`	`plug_line_a`

bus_1	plug_bus	load_1	plug_load
bus_1	plug_bus	hp	plug_load
bus_1	plug_bus	pv_1	plug_pv
bus_2	plug_bus	line_2	plug_line_b
bus_2	plug_bus	load_2	plug_load
bus_2	plug_bus	pv_2	plug_pv
ext_th_grid	flange_supply	pipe_1_supply	flange_a
ext_th_grid	flange_return	pipe_1_return	flange_b
junction_tank_supply	flange_1	pipe_1_supply	flange_b
junction_tank_supply	flange_2	pipe_tank_supply	flange_b
junction_tank_supply	flange_3	pipe_2_supply	flange_a
junction_tank_return	flange_1	pipe_2_return	flange_b
junction_tank_return	flange_2	pipe_tank_return	flange_a
junction_tank_return	flange_3	hp	flange_evap_in
junction_1_supply	flange_1	pipe_2_supply	flange_b
junction_1_supply	flange_2	pipe_3_supply	flange_a
junction_1_supply	flange_3	pipe_4_supply	flange_a
junction_1_return	flange_1	pipe_4_return	flange_b
junction_1_return	flange_2	pipe_3_return	flange_b
junction_1_return	flange_3	pipe_2_return	flange_a
junction_2_supply	flange_1	pipe_4_supply	flange_b
junction_2_supply	flange_2	pipe_5_supply	flange_a
junction_2_supply	flange_3	pipe_6_supply	flange_a
junction_2_return	flange_1	pipe_6_return	flange_b
junction_2_return	flange_2	pipe_5_return	flange_b
junction_2_return	flange_3	pipe_4_return	flange_a

hex_1	flange_supply	pipe_3_supply	flange_b
hex_1	flange_return	pipe_3_return	flange_a
hex_2	flange_supply	pipe_5_supply	flange_b
hex_2	flange_return	pipe_5_return	flange_a
hp	flange_evap_out	pipe_1_return	flange_a
hp	flange_cond_in	pump_cond	flange_out
hp	flange_cond_out	tank	flange_hp_in
hp	meas_p_el	flex_heat_ctrl	meas_hp_p_el
tank	meas_temp	flex_heat_ctrl	meas_tank_temp
tank	flange_hp_out	pump_cond	flange_in
tank	flange_tank_return	pipe_tank_return	flange_b
tank	flange_tank_supply	pump_tank	flange_in
pump_tank	flange_out	pipe_tank_supply	flange_a
bypass	flange_supply	pipe_6_supply	flange_b
bypass	flange_return	pipe_6_return	flange_a
voltage_ctrl	meas_voltage_pu	bus_1	meas_voltage_pu
voltage_ctrl	setpoint_hp_p_el	flex_heat_ctrl	setpoint_hp_p_el
flex_heat_ctrl	setpoint_tank_mflow	pump_tank	setpoint_mflow
flex_heat_ctrl	setpoint_cond_mflow	pump_cond	setpoint_mflow